

In [10]:

```

try:
    a=10
    b=0
    print(a/b)
except ZeroDivisionError:
    print("Zero Divison Error Rised")
    print(a/(b+0.1))

```

Zero Divison Error Rised
100.0

In [13]:

```

# default except is a except block without any ExceptionClass name.
#it must be at last of all except blocks
try:
    a=10
    b='sa'
    print(a/b)
except ZeroDivisionError:
    print("Zero Divison Error Rised")
    print(a/(b+0.1))
except:
    print("this is default except block , please provide some number")

```

this is default except block , please provide some number

In [14]:

```

# single except block that can handle multiple Exceptions
try:
    a=10
    b=0
    print(a/b)
except (ZeroDivisionError, ValueError, ArithmeticError): #as msg alias
    print("Zero Divison Error Rised")
    print(a/(b+0.1))

```

Zero Divison Error Rised
100.0

In [18]:

```

try:
    a=10
    b=0
    print(a/b)
except ZeroDvdsError:
    print("Zero Divison Error Rised")
    print(a/(b+0.1))
finally:
    print("finally")

```

Out[18]:

```

'try:\n    a=10\n    b=0\n    print(a/b)\nexcept ZeroDvdsError:\n    print\n("Zero Divison Error Rised")\n    print(a/(b+0.1))\nfinally:\n    print("fin\nally") '

```

In []:

```
import os
try:
    a=10
    b=0
    print('try')
    os._exit(0)
    print(a/b)
except ZeroDivisionError:
    print("Zero Divison Error Rised")
    print(a/(b+0.1))
finally:
    print("finally")
```

In [3]:

```
#else block excuted if no errors in try block
try:
    a=10
    b=1
    print(a/b)
except ZeroDivisionError:
    print("Zero Divison Error Rised")
    print(a/(b+0.1))
else:
    print('else block')
finally:
    print("finally")
```

```
10.0
else block
finally
```

In [54]:

```
# user defined exception , all exceptions are directly or indirectly inherited from BaseExc

class TooYoungException(BaseException):
    def __init__(self,msg):
        msg="please provide valid age"
        print(msg)

class TooOldException(BaseException):
    def __init__(self,msg):
        msg="please provide valid age"
        print(msg)

age=3
if age<15:
    raise TooYoungException("you are too young")
elif age>60:
    raise TooOldException("you are too old")
else:
    print(age)
```

please provide valid age

```
-----
TooYoungExceptionTraceback (most recent call last)
Input In [54], in <cell line: 13>()
    12 age=3
    13 if age<15:
--> 14     raise TooYoungException("you are too young")
    15 elif age>60:
    16     raise TooOldException("you are too old")
```

TooYoungException: you are too young

formatting the strings

In [60]:

```
name = "python"
year = 1991
rank = 1
print("{} was published in {} , it's rank is {}".format(name,year,rank))    # default argum
print("{2} was published in {1} , it's rank is {0}".format(rank,year,name))  # positional
print("{x} was published in {y} , it's rank is {z}".format(x=name,y=year,z=rank))  # keywo
print("Thi float number is : {:.2}".format("python"))
```

```
python was published in 1991 , it's rank is 1
python was published in 1991 , it's rank is 1
python was published in 1991 , it's rank is 1
Thi float number is : py
```

formatting numbers

In [58]:

```
a="{:b}".format(5)
a
```

Out[58]:

```
'101'
```

In [30]:

```
print("Thi integer number is : {}".format(15))
print("Thi integer number is : {:d}".format(15))
print("Thi integer number is : {:o}".format(64))    #octal number
print("Thi integer number is : {:010d}".format(123)) #Length=10,filled with 0's and right
print("Thi integer number is : {:X}".format(123))
print("Thi integer number is : {:b}".format(15))
```

```
Thi integer number is : 15
Thi integer number is : 15
Thi integer number is : 100
Thi integer number is : 0000000123
Thi integer number is : 7B
Thi integer number is : 1111
```

formatting floating numbers

In [42]:

```
print("Thi float number is : {}".format(123.4567))
print("Thi float number is : {:.f}".format(123.4567))
print("Thi float number is : {:018f}".format(123.4567))    # Length=18 , filled with 0's
print("Thi float number is : {:.2f}".format(123.4567))
print("Thi float number is : {:018.2f}".format(123.4567))
```

```
Thi float number is : 123.4567
Thi float number is : 123.456700
Thi float number is : 00000000123.456700
Thi float number is : 123.46
Thi float number is : 000000000000123.46
```

In [59]:

```
# dictionary formatting
person={'year':1991,'name':'python'}
print("{p[name]} was published in {p[year]}".format(p=person))
print("{name} was published in {year}".format(**person))
```

```
python was published in 1991
python was published in 1991
```

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